

Tutorial Sheet

B.Tech.(C.S.)

Session 2024-25

Subject : Theory of Automata and Formal Language

Course Code:BCSC0011

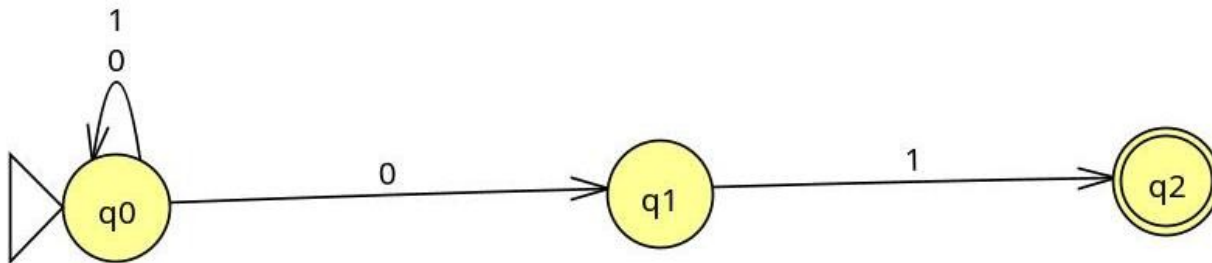
Deterministic Finite Automaton Question:

1. Construct a DFA over the alphabet $\Sigma = \{a,b\}$ accepting the string starts with ab .
2. Construct a DFA over the alphabet $\Sigma = \{a,b\}$ such that length of the string is at least 2.
3. Construct a DFA over the alphabet $\Sigma = \{a,b\}$ such that length of the string is 2.
4. Construct a DFA over the alphabet $\Sigma = \{a,b\}$ such that length of the string is less than 2.
5. Construct a DFA over $w = \{a,b\}^*$ where $|w| \bmod 3 = 0$.
6. Construct a DFA over $\Sigma = \{a,b\}$ where $n_a(w) = 2$.
7. Construct a DFA over $\Sigma = \{a,b\}$ where $n_a(w) \bmod 2 = 0$.
8. Construct a DFA over alphabet $\Sigma = \{a,b\}$ where $n_a(w) = 2$ and $n_b(w) = 2$.
9. Construct a DFA over the alphabet $\Sigma = \{0,1\}$ which interpreted as binary number is divisible by 3.
10. Construct a DFA over the alphabet $\Sigma = \{a,b\}$ and all the strings start with a.
11. Construct a DFA over the alphabet $\Sigma = \{a,b\}$ and all the strings contains a.
12. Construct a DFA over the alphabet $\Sigma = \{a,b\}$ and all the strings ends with a.
13. Construct a DFA over the alphabet $\Sigma = \{a,b\}$ and all the strings start with ab.
14. Construct a DFA over the alphabet $\Sigma = \{a,b\}$ and all the strings containing ab.
15. Construct a DFA over the alphabet $\Sigma = \{a,b\}$ and all the strings ends with ab.
16. Construct a DFA over the alphabet $\Sigma = \{a,b\}$ and string start with a and end with b.
17. Construct a DFA over the alphabet $\Sigma = \{a,b\}$ and strings start with different symbols.
18. Construct a DFA over the alphabet $\Sigma = \{a,b\}$ and strings start and end with same symbol.
19. Construct a DFA over alphabet $\Sigma = \{a,b\}$ where every a is followed by b.
20. Construct a DFA over alphabet $\Sigma = \{a,b\}$ where every a is not followed by b.
21. Construct a DFA over alphabet $\Sigma = \{a,b\}$ where every a has to be followed by bb.
22. Construct a DFA over alphabet $\Sigma = \{a,b\}$ where every a must not be followed by bb.
23. Let $\Sigma = \{a, b\}$ and let $L = \{a^n b^m \mid n, m \geq 1\}$. Design a DFA for L.
24. Let $\Sigma = \{a, b, c\}$ and let $L = \{a^n b^m c^l \mid n, m, l \geq 1\}$. Design a DFA for L.
25. Let $\Sigma = \{a, b, c\}$ and let $L = \{a^n b^m c^l \mid n, m, l \geq 0\}$. Design a DFA for L.
26. Construct a DFA over the alphabet $\Sigma = \{a,b\}$ such that the second symbol from the LHS is a.
27. Construct a DFA over $\Sigma = \{a\}$ such that $\{a^n \mid n \geq 0, n \neq 3\}$.
28. Construct a DFA over the alphabet Let $\Sigma = \{a, b\}$ accepting the string start and end with same symbol .
29. Design a DFA for the language $L = \{w : n_a = 1, w \in (a,b)^*\}$.
30. Design a DFA for the language $L = \{w : n_a(w) \geq 1, w \in (a,b)^*\}$.
31. Design a DFA for the language $L = \{w : n_a(w) \leq 3, w \in (a,b)^*\}$.

32. Design a DFA for the language ,contains strings in which left most symbol differ from the right most symbol. Σ is given $\{0,1\}$.
33. Design a DFA which accepts set of strings such that every string containing 00 as a substring but not 000 as a substring.

Non Deterministic Finite Automaton Questions:

1. Design an NFA over the alphabet $\Sigma=\{a,b\}$ which accepts all strings start with a.
2. Design an NFA over $\Sigma=\{a,b\}$ which contains a.
3. Design an NFA over $\Sigma=\{a,b\}$ which ends a.
4. Design an NFA over $\Sigma=\{a,b\}$ which starts with ab.
5. Design an NFA over $\Sigma=\{a,b\}$ which contains ab.
6. Design an NFA over $\Sigma=\{a,b\}$ which ends with ab.
7. Design an NFA accepting all strings over $\{0,1\}$ which ends with 1 but does not contain the substring 00.
8. Design an NFA which accepts language $L=\{w \in (a,b)^* | w \geq 3 \text{ and third symbol of } w \text{ from right is } a\}$.
9. Design an NFA over the alphabet $\Sigma=\{a,b\}$ and all the strings start with ab.
10. Design an NFA over the alphabet $\Sigma=\{a,b\}$ and all the strings ends with ab.
11. Design an NFA over the alphabet $\Sigma=\{a,b\}$ and all the strings containing ab.
12. Design a NFA for the language $L=\text{all strings over } \{0,1\} \text{ that have at least two consecutive 0,s or 1,s.}$
13. Consider NFA in the following figure describe the processing of string 00101 by it.



14. Let $M = (\{q1, q2, q3\}, \{0,1\}, \delta, \{q1\}, \{q3\})$ is a NFA where δ is given as :

$$\delta(q1, 0) = \{q1, q3\}$$

$$\delta(q1, 1) = \{q1\}$$

$$\delta(q2, 0) = \{q1, q2\}$$

$$\delta(q2, 1) = \{q1, q2\}$$

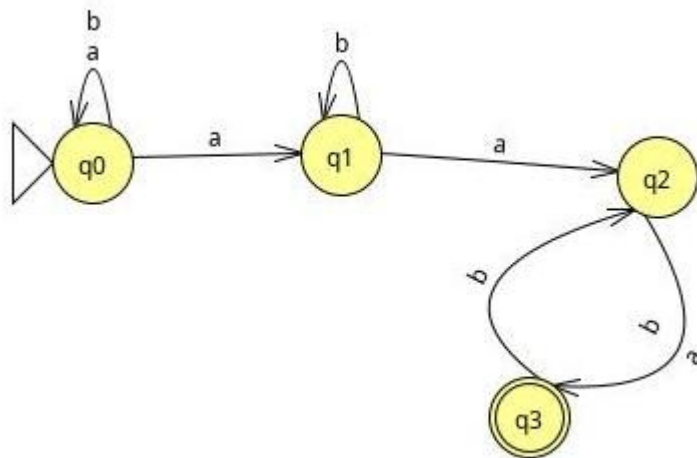
$$\delta(q3, 1) = \{q1, q2\}$$

construct the transition diagram corresponding to NFA.

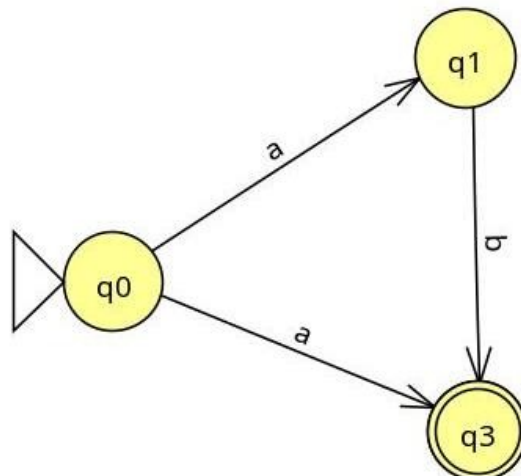
15. design a NFA accepting all strings over $\{0,1\}$ which ends in 1 but does not contain the substring 00.

NFA to DFA Conversion Questions:

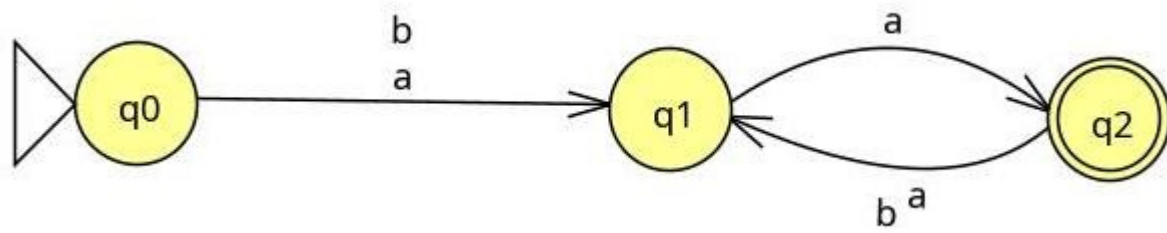
1. Design a NFA over the alphabet $\{0,1\}$ which end with 00 and convert it to the DFA.
2. Design a NFA over the alphabet $\{a,b\}$ which starts with a and convert it to the DFA.
3. Design a NFA over the alphabet $\{a,b\}$ which accepts the strings ending with a and convert it into the DFA.
4. Design a NFA over the alphabet $\{a,b\}$ which accepts the strings in which the second symbol from RHS is a, Convert it to DFA
5. Design a NFA over the alphabet $\{a,b\}$ which accepts the strings in which the third symbol from RHS is a, Convert it to DFA.
6. Convert the following NFA in to DFA.



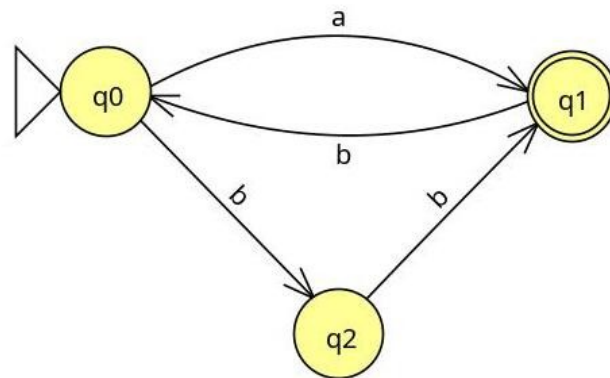
7. Convert the following NFA in to DFA.



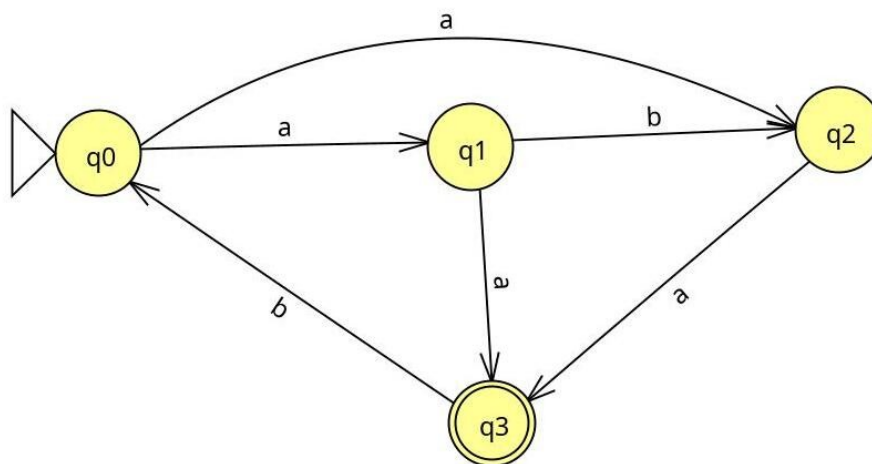
8. Convert the following NFA to DFA.



9. Convert the following NFA to DFA.

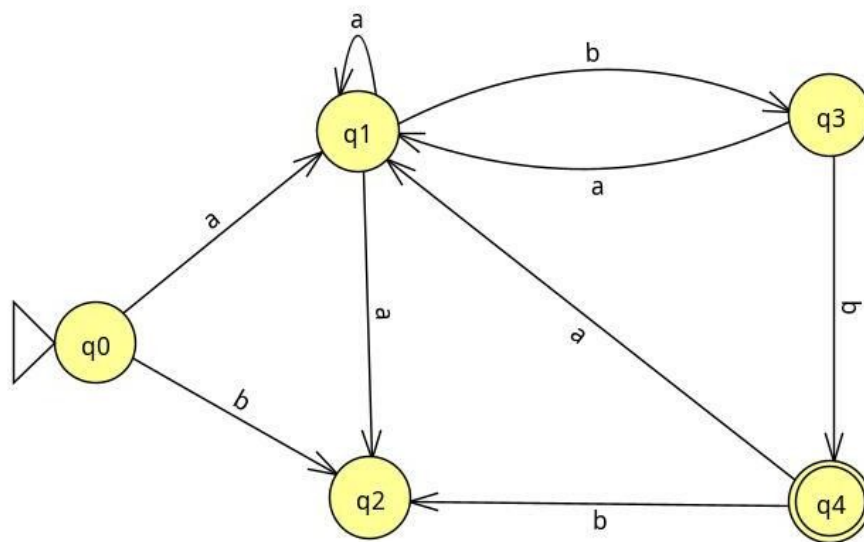


10. Convert the following NFA to DFA.

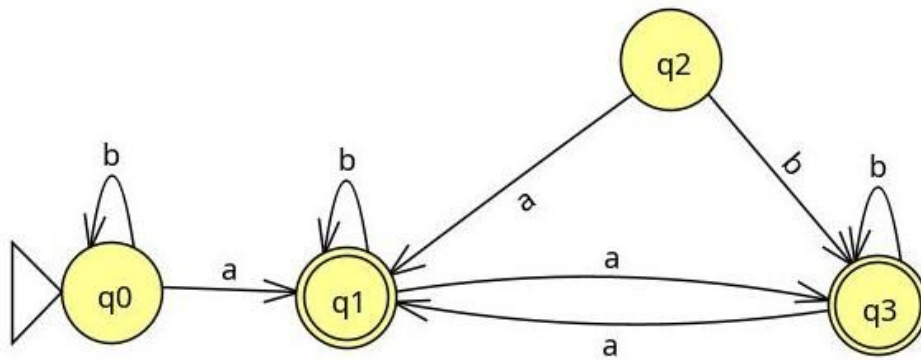


DFA Minimization:

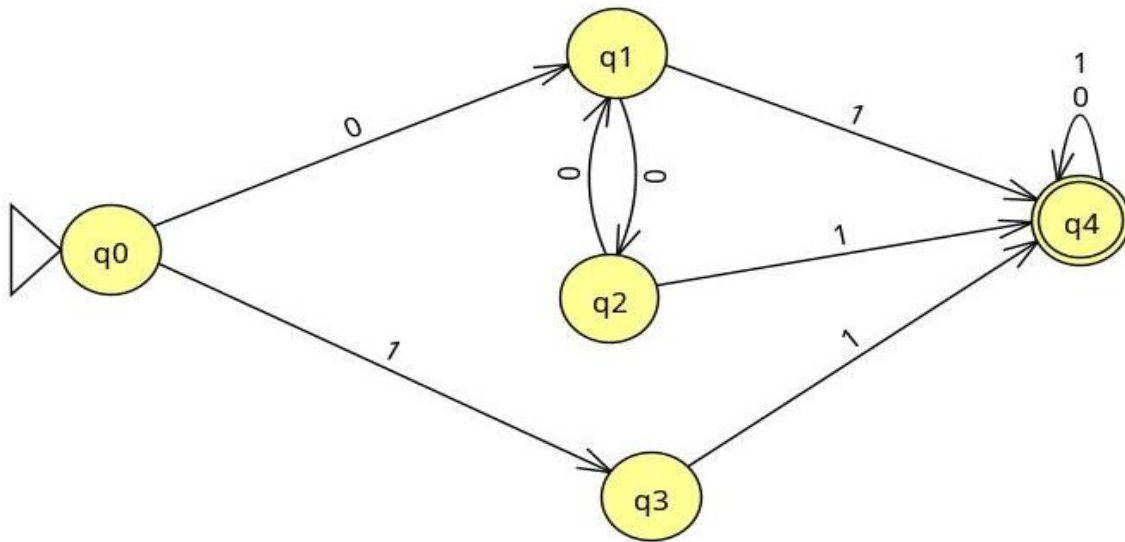
1. Minimize the following DFA.



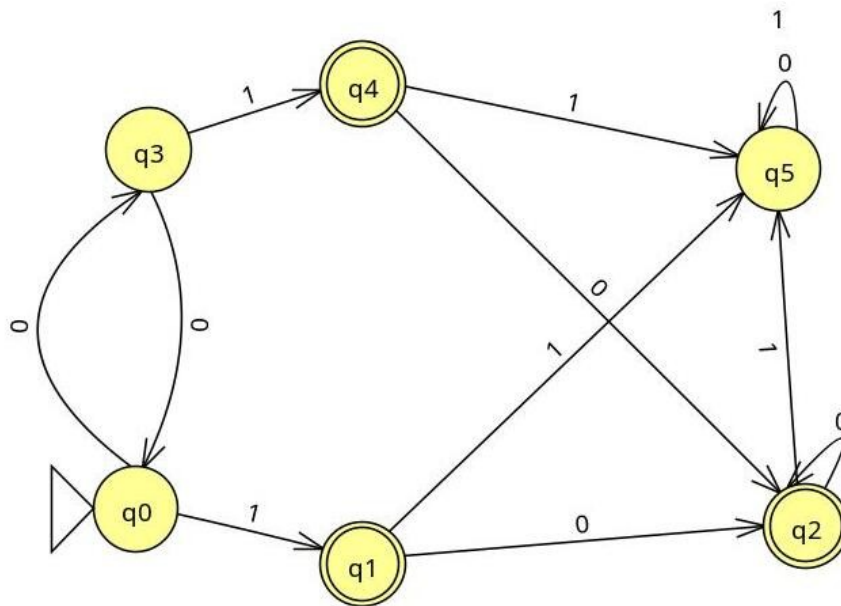
2. Minimize the following DFA.



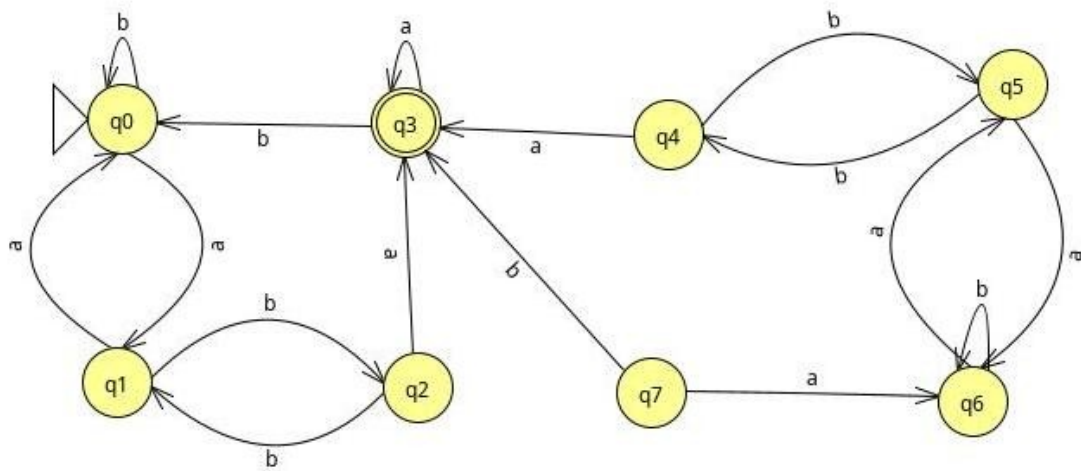
3. Minimize the following DFA.



4. Minimize the following DFA.

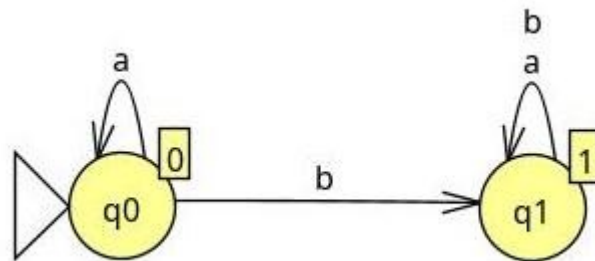


5. Minimize the following DFA.

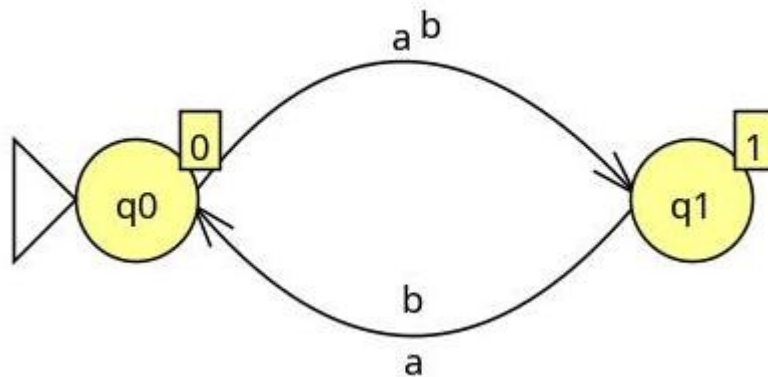


Moore and Mealy Machine:

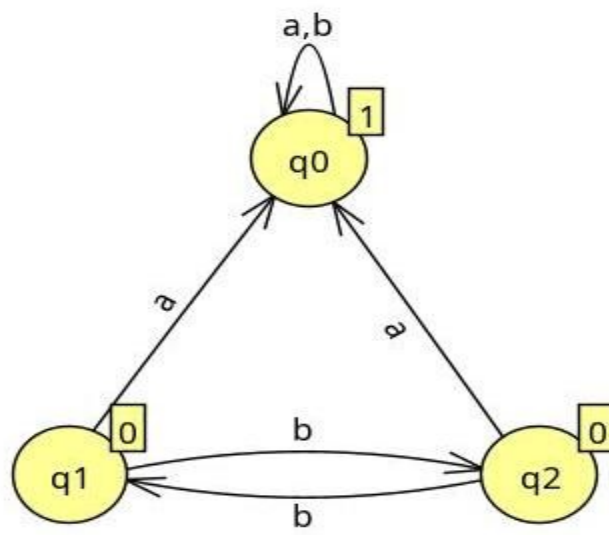
1. Design a Moore machine which counts the occurrence of substring aab in input string.
2. Design a Mealy machine, which prints 1, s complement of input bit string over the alphabet $\Sigma=\{0,1\}$.
3. Convert the following Moore machine into Mealy Machine.



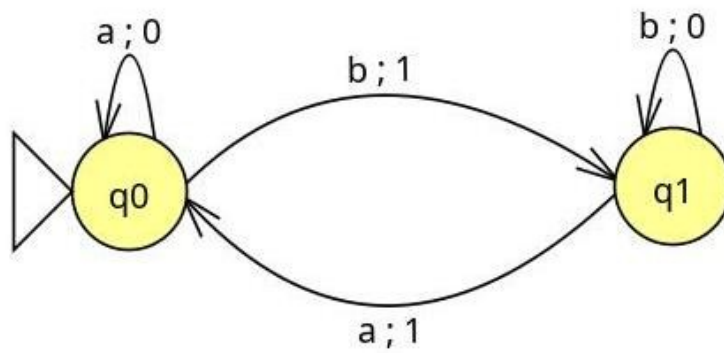
4. Convert the following Moore machine into Mealy Machine.



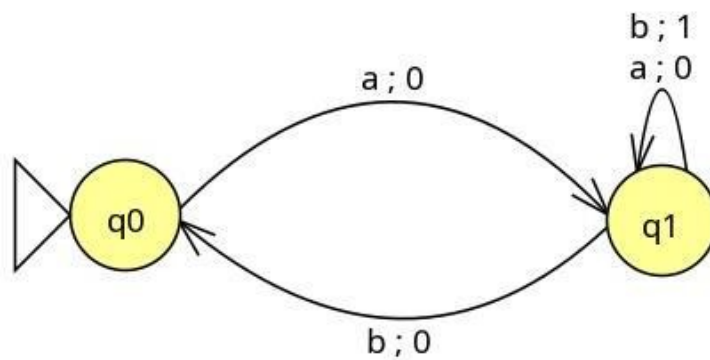
5. Convert the following Moore machine into Mealy Machine.



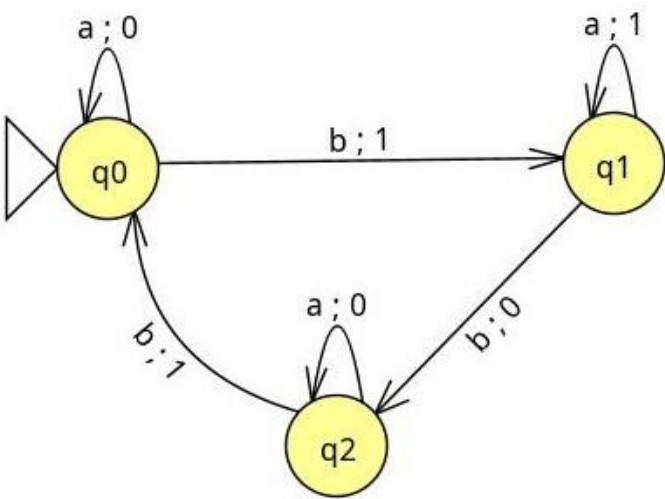
6. Convert the following Mealy machine into Moore machine.



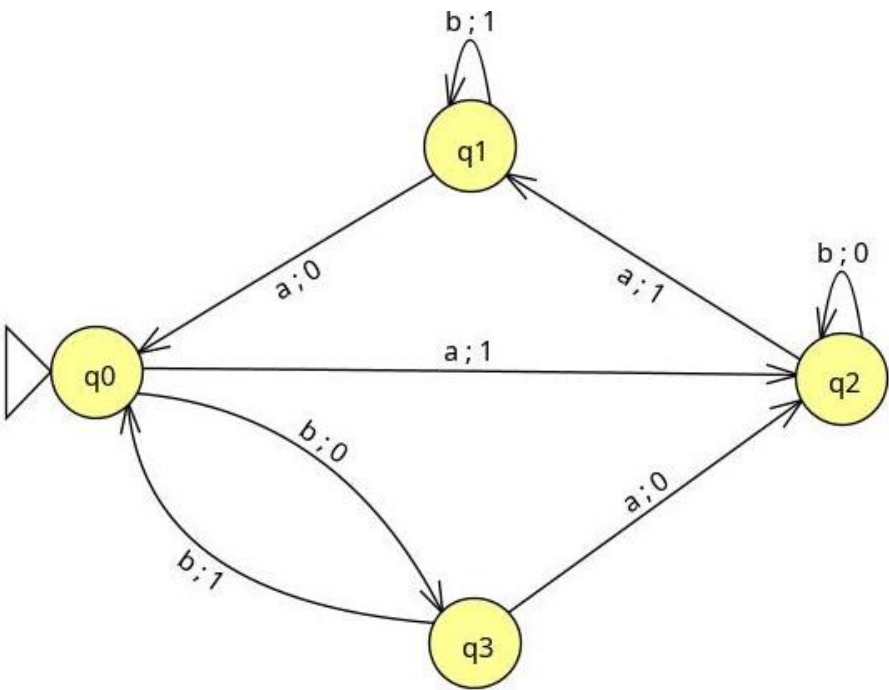
7. Convert the following Mealy machine into Moore machine.



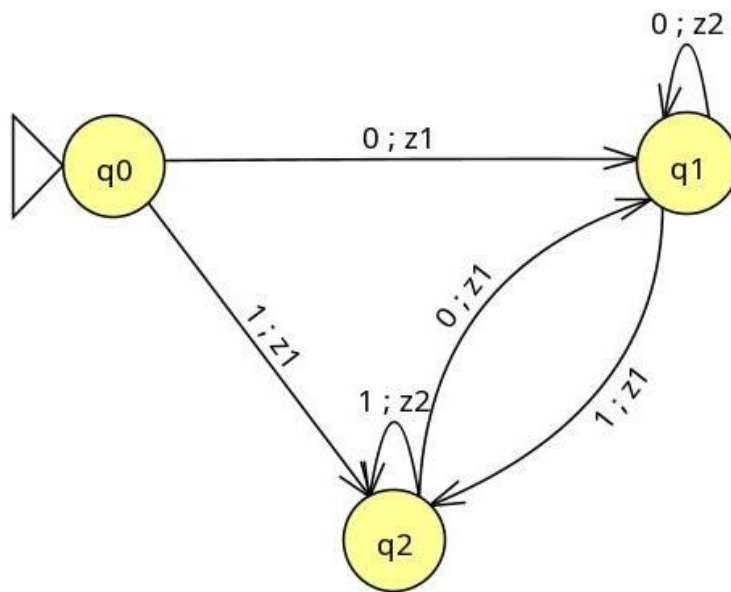
8.Convert the following Mealy machine into Moore machine.



9.Convert the following Mealy machine into Moore machine.



10.Convert the following Mealy machine into Moore machine.



Regular Expression :

- Write the regular expression over alphabet (a,b,c) containing at least one a and at least one b.
- Write the regular expression for the set of strings of 0,s and 1,s whose tenth symbol from the right end is 1.
- Write a regular expression for the set of strings of an equal number of 0,s and 1,s such that ,the numbers of 0,s differ from the number of 1,s by at most 1.
- Write a regular expression for the set of strings of 0,s and 1,s not containing 101 as a sub string.
- Write the regular expression for the language
 $L = \{ab^n \mid n \geq 3, w \in (a,b)^+\}$.
- Write the regular expression for the language
 $L = \{w \mid |w| \bmod 3 = 0\}, w \in (a,b)^*$.
- Find the regular expression for the following
 - $L = \{a^n b^m \mid n \geq 1, m \geq 1, n, m \geq 3\}$
 - $L = \{ab^n w \mid n \geq 3, w \in (a,b)^+\}$
 - $L = \{a^n \mid n \geq 0, n \neq 3\}$
- Write the regular expression for the following language over the alphabet {0,1}
 - All strings ending in 01.
 - All strings not ending in 01.
 - All strings containing an even number of 0,s.
 - All strings with at most two occurrence of substring 00.
 - All strings not containing the substring 10.
- Construct the finite automata for the regular expression
 $r = (a+b)^* abb$.
- Prove that
 - $(r_1^*)^* = r_1^*$
 - $r_1^* (r_1 + r_2)^* = (r_1 + r_2)^*$
 - $(r_1 + r_2)^* = (r_1^* r_2^*)^*$